# MSG No. 182.182 $P6'_322'$ [ Type III, hexagonal ]

Table 1: Wyckoff site: `2a`, site symmetry: `32.`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ 0,\ 0]$ | `[1,2,3,4,5,6]` |
| 2 | $\left[0,\ 0,\ \frac{1}{2}\right]$ | `[7,8,9,10,11,12]` |

Table 2: Wyckoff site: `2b`, site symmetry: `3.2'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ 0,\ \frac{1}{4}\right]$ | `[1,2,3,10,11,12]` |
| 2 | $\left[0,\ 0,\ \frac{3}{4}\right]$ | `[4,5,6,7,8,9]` |

Table 3: Wyckoff site: `2c`, site symmetry: `3.2'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$ | `[1,2,3,10,11,12]` |
| 2 | $\left[\frac{2}{3},\ \frac{1}{3},\ \frac{3}{4}\right]$ | `[4,5,6,7,8,9]` |

Table 4: Wyckoff site: `2d`, site symmetry: `3.2'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$ | `[1,2,3,10,11,12]` |
| 2 | $\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$ | `[4,5,6,7,8,9]` |

Table 5: Wyckoff site: `4e`, site symmetry: `3..`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ 0,\ z]$ | `[1,2,3]` |
| 2 | $[0,\ 0,\ -z]$ | `[4,5,6]` |
| 3 | $\left[0,\ 0,\ z+\frac{1}{2}\right]$ | `[7,8,9]` |
| 4 | $\left[0,\ 0,\ \frac{1}{2}-z\right]$ | `[10,11,12]` |

Table 6: Wyckoff site: `4f`, site symmetry: `3..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$ | `[1,2,3]` |
| 2 | $\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$ | `[4,5,6]` |
| 3 | $\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$ | `[7,8,9]` |
| 4 | $\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}-z\right]$ | `[10,11,12]` |

Table 7: Wyckoff site: `6g`, site symmetry: `.2.`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ 0,\ 0]$ | `[1,4]` |
| 2 | $[0,\ x,\ 0]$ | `[2,5]` |
| 3 | $[-x,\ -x,\ 0]$ | `[3,6]` |
| 4 | $\left[x,\ x,\ \frac{1}{2}\right]$ | `[7,10]` |
| 5 | $\left[-x,\ 0,\ \frac{1}{2}\right]$ | `[8,11]` |
| 6 | $\left[0,\ -x,\ \frac{1}{2}\right]$ | `[9,12]` |

Table 8: Wyckoff site: `6h`, site symmetry: `..2'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ 2x,\ \frac{1}{4}\right]$ | `[1,11]` |
| 2 | $\left[-2x,\ -x,\ \frac{1}{4}\right]$ | `[2,12]` |
| 3 | $\left[x,\ -x,\ \frac{1}{4}\right]$ | `[3,10]` |
| 4 | $\left[-x,\ -2x,\ \frac{3}{4}\right]$ | `[4,8]` |
| 5 | $\left[2x,\ x,\ \frac{3}{4}\right]$ | `[5,9]` |
| 6 | $\left[-x,\ x,\ \frac{3}{4}\right]$ | `[6,7]` |

Table 9: Wyckoff site: `12i`, site symmetry: `1`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ z]$ | `[1]` |
| 2 | $[-y,\ x-y,\ z]$ | `[2]` |
| 3 | $[-x+y,\ -x,\ z]$ | `[3]` |
| 4 | $[x-y,\ -y,\ -z]$ | `[4]` |
| 5 | $[y,\ x,\ -z]$ | `[5]` |
| 6 | $[-x,\ -x+y,\ -z]$ | `[6]` |
| 7 | $\left[x-y,\ x,\ z+\frac{1}{2}\right]$ | `[7]` |
| 8 | $\left[-x,\ -y,\ z+\frac{1}{2}\right]$ | `[8]` |
| 9 | $\left[y,\ -x+y,\ z+\frac{1}{2}\right]$ | `[9]` |
| 10 | $\left[x,\ x-y,\ \frac{1}{2}-z\right]$ | `[10]` |
| 11 | $\left[-x+y,\ y,\ \frac{1}{2}-z\right]$ | `[11]` |
| 12 | $\left[-y,\ -x,\ \frac{1}{2}-z\right]$ | `[12]` |